

ATTACHMENT B

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EXECUTIVE SUMMARY

HHS’s RFI recognizes that AI adoption in clinical care is constrained not only by technology but by the rules, incentives, and operational infrastructure that govern care delivery. The RFI explicitly frames HHS’s levers as regulation, reimbursement, and research & development—and seeks concrete, experience-based input from builders, buyers, evaluators, users, and recipients of AI-enabled care. [FR-1]

This comment advances a system-design position: the central challenge for clinical AI at scale is governance and incentive alignment, not model capability alone—particularly if AI is to reduce administrative redundancy, consolidate longitudinal health information (including biologic, social, and structural determinants), and improve continuity of care at scale. HHS’s AI Strategy similarly emphasizes governance and risk management, public trust, transparency, accountability, and respect for Americans’ rights and protections associated with health information. With aligned governance, AI can reduce duplicative documentation and testing while improving longitudinal continuity; without it, AI risks amplifying fragmentation and administrative burden. [HHS-AI-1]

To clarify roles and failure modes, this comment uses three operational definitions and recommends that HHS prioritize (1) a reimbursable pathway for evaluation and implementation work, (2) “AI-ready interoperability and security,” (3) governance expectations for non-device AI, (4) anti-extraction safeguards, and (5) an evidence agenda emphasizing implementation science and post-deployment system behavior. [FR-1]

PURPOSE AND FRAMING

HHS’s RFI seeks to accelerate adoption and use of AI “as part of clinical care,” while fostering public trust, reducing uncertainty that impedes innovation, and aligning incentives so AI reduces burden and improves outcomes. The RFI also notes that HHS divisions—including FDA, NIH, CMS, and ASTP/ONC—have recently sought public feedback on AI-related issues, and this RFI is intended to bring those threads into a coordinated posture. [FR-1]

This response is intentionally neutral and non-promotional. It does not propose a specific commercial product. Instead, it describes conditions under which AI can operate responsibly at scale—particularly for longitudinal and agentic systems—without compromising privacy, trust, equity, or clinician judgment. [FR-1]

OPERATIONAL DEFINITIONS

Health Principal: the individual whose health, well-being, and health data are implicated across the lifespan, regardless of episodic contact with the health system. This term distinguishes longitudinal agency from the episodic framing often implied by “patient” and aligns with federal policy emphasizing enforceable access rights and patient-directed data exchange (e.g., HIPAA access rights; CMS patient access APIs). [HIPAA-ACCESS-1] [CMS-API-1]

Unusable Friction: governance, consent, or oversight requirements that are individually well-intentioned but—when applied transaction-by-transaction or without automation—render responsible systems infeasible at scale, thereby restricting access and exacerbating inequity, particularly in under-resourced settings. This concept underscores the need for lifecycle evaluation and continuous monitoring rather than episodic or manual oversight models. [NIST-1]

In the loop / on the loop: the Health Principal is “in the loop” because their intent, values, and permissions should guide system behavior longitudinally; clinicians and other professionals are “on the loop,” engaged through risk-tiered oversight, escalation pathways, and accountability mechanisms rather than embedded in every routine interaction. This distinction supports scalable safety controls consistent with lifecycle governance approaches reflected in NIST and FDA/IMDRF lifecycle framing. [NIST-1] [FDA-GMLP-1]

PRIORITIES ACROSS THE RFI QUESTIONS

Across the RFI questions, this comment recommends that HHS prioritize:

1. A reimbursable pathway for AI evaluation and implementation work (including silent-mode testing and structured post-deployment monitoring);
2. “AI-ready interoperability and security” capabilities (provenance, audit logs, version control, secure access controls, adversarial resilience, and computable permissions);
3. Clear governance and cybersecurity expectations for non-medical-device AI used in clinical operations;
4. Anti-extraction and anti-capture safeguards to prevent vendor lock-in, data hoarding, and assurance regimes that entrench incumbency;
5. An evidence agenda emphasizing implementation science, human factors, equity monitoring, security performance, and post-deployment system behavior. [FR-1]

This comment includes a one-page Pilot Invitation outlining an IRB-governed, phased, real-world evaluation of a longitudinal, multi-agent AI support model (non-branded), with explicit commitments to patient-directed permissions, security safeguards, and tribal data stewardship. [45CFR46-1]

CONTEXT: WHY GOVERNANCE MATTERS NOW

The diagnostic and care delivery system’s baseline error burden is substantial. A National Academies report states that a conservative estimate found that 5% of U.S. adults who seek outpatient care each year experience a diagnostic error, and diagnostic errors contribute to significant harm. Meanwhile, major national analyses have identified clinician burnout and administrative burden as threats to quality and professional well-being, reinforcing the need for systems that reduce unusable friction rather than adding to it. [NASEM-1] [NAM-1]

AI will not automatically resolve these systemic problems. Evidence shows that AI systems can match or exceed clinician performance in certain constrained tasks, but real-world outcomes depend on workflow integration, escalation pathways, and governance design. For example, a randomized clinical trial found that physicians with access to an LLM did not significantly improve diagnostic reasoning compared with conventional resources, even though the LLM alone scored higher. A 2025 systematic review and meta-analysis comparing generative AI with physicians in diagnostic tasks found no significant overall difference in aggregate performance and worse performance compared with expert physicians, supporting a cautious, risk-tiered implementation approach. [GOH-1] [TAKITA-1]

The relevant policy question is therefore not whether AI can answer a diagnostic prompt, but whether it can be governed to reduce preventable health loss across time—while preserving safety, accountability, and equity. Without aligned governance and incentive structures, AI risks automating fragmentation and administrative burden; with them, it may help reduce delay,

strengthen continuity, and mitigate the cumulative harms embedded in current workflows. [NIST-1] [WHO-LMM-1]

OVERARCHING RECOMMENDATIONS ALIGNED TO HHS'S STATED LEVERS

HHS should use the RFI's three levers to create a coherent and scalable adoption environment:

Regulation: codify scalable governance and cybersecurity expectations for non-device clinical AI; define risk-tiered oversight models; strengthen interoperability requirements for provenance, auditability, and secure access controls; and clarify privacy guidance for AI-enabled clinical and operational use cases. [ONC-DSI-1] [ONC-USCDI-1] [IB-1] [HIPAA-TPO-1]

Reimbursement: establish payment pathways that support real-world evaluation, phased implementation, and structured post-deployment monitoring—similar to how CMS recognizes non-face-to-face care management and clinical complexity within existing payment policy. [CMS-CCM-1] [PFS-STAT-1] [PFS-REG-1]

Research & development: fund implementation science and shared evaluation infrastructure; support multi-site learning networks and lifecycle monitoring toolkits; and prioritize equity, human factors, and security performance. AHRQ's CDSiC provides a model for incorporating clinicians, patients, and advocates into evaluation design and resource development. [AHRQ-CDSIC-1]

QUESTION 1

What are the biggest barriers to private sector innovation in AI for health care and its adoption and use in clinical care?

The largest barriers to innovation and adoption are structural, economic, and governance-related:

Misaligned incentives under episodic payment models. Many clinically valuable AI interventions are deflationary: they reduce avoidable utilization, administrative burden, and capacity strain. Entities bearing longitudinal risk are more naturally incentivized to promote high-value adoption, whereas fee-for-service payment structures may delay coverage and create unintended spending distortions. [FR-1]

Fragmented and uneven interoperability implementation. Longitudinal AI depends on consistent, standards-based access to data elements (USCDI) and interoperable APIs (FHIR) that function reliably across settings. Although federal policy has established these foundations through ONC certification criteria and CMS interoperability rules, implementation remains uneven and costly, increasing integration friction and favoring incumbent vendors. [ONC-USCDI-1] [ONC-API-1]

Workflow integration, human factors, and embedded bias risk. Even high-performing AI systems can fail if they increase cognitive workload, exacerbate alert fatigue, or introduce new failure modes such as automation bias. Adoption must therefore include human factors evaluation, equity monitoring, and lifecycle governance—not accuracy metrics alone. [NIST-1]

Unclear accountability for non-medical-device AI. Many AI systems influencing care (e.g., documentation support, triage routing, care navigation) fall outside traditional medical device pathways. ONC's Predictive Decision Support Intervention (Predictive DSI) requirements introduce transparency and intervention risk management expectations, but operational AI uses remain inconsistently governed, increasing procurement uncertainty and slowing adoption. [ONC-DSI-1]

Absence of standardized lifecycle evaluation for non-device AI. Regulatory frameworks increasingly emphasize lifecycle management, including post-deployment monitoring and change

management. However, standardized expectations for drift detection, performance monitoring, cybersecurity resilience, and incident response remain uneven for non-device AI, requiring individual health systems to build bespoke oversight infrastructure and raising barriers to diffusion. [NIST-1] [FDA-GMLP-1]

Safety, security, and trust considerations in the use of general-purpose AI. As access gaps, administrative burden, and unusable friction persist within the delivery system, individuals increasingly turn to general-purpose AI tools to seek information and guidance. This behavior reflects unmet need rather than misconduct. The policy concern is not the exercise of individual autonomy, but the absence of governance frameworks that ensure transparency, validation standards, escalation pathways, documentation integrity, and security safeguards when AI influences health-related decisions. [ECRI-1] [WHO-LMM-2]

These barriers intersect with existing statutory and regulatory authorities governing payment policy, interoperability and certification, privacy protections, and information blocking enforcement (see, e.g., 42 U.S.C. 1395w-4; 42 CFR parts 410, 414, 425, and 438; 45 CFR parts 160, 164, 170, and 171). [PFS-STAT-1] [PFS-REG-1] [ONC-USCDI-1] [IB-1] [HIPAA-160-1] [MSSP-1] [MMC-1]

QUESTION 2

What regulatory, payment policy, or programmatic design changes should HHS prioritize to incentivize the effective use of AI in clinical care and why? Provide specific changes and applicable CFR citations.

HHS should prioritize reforms that align incentives with outcomes, longitudinal continuity, access, equity, privacy, accountability, and individual autonomy and responsibility—rather than volume or administrative throughput. The following priority actions would create a coherent adoption environment that supports trustworthy collaboration among individuals, clinicians, and AI-enabled systems. [FR-1]

Applicable authorities (valuation and physician fee schedule anchors): 42 U.S.C. 1395w-4; 42 CFR part 414. [PFS-STAT-1] [PFS-REG-1]

1. Establish a reimbursable pathway for AI evaluation, implementation, and lifecycle monitoring.
Adoption is constrained not by model availability, but by the absence of reimbursement for responsible deployment activities: silent-mode testing, workflow redesign, staff training, monitoring infrastructure, equity assessment, cybersecurity safeguards, and incident response processes. HHS should create a time-limited, reimbursable implementation pathway modeled on existing non-face-to-face care management and longitudinal complexity payment structures, while explicitly recognizing safety and monitoring work as clinical infrastructure.
Applicable authorities: 42 CFR part 410; 42 CFR part 414. [PFS-410-1] [PFS-REG-1] [CMS-CCM-1]
2. Modernize valuation methodologies and strengthen transparency in the representation of longitudinal value.
Current valuation methodologies are structurally organized around discrete services and procedures. Advisory processes that inform RVU development have historically focused on time, intensity, and procedural work, which can limit representation of longitudinal prevention, care coordination, stabilization, and infrastructure that preserves clinical

capacity. As a result, activities that reduce avoidable deterioration or downstream utilization are under-recognized in payment design. Because CMS retains ultimate statutory authority over valuation and payment policy, HHS should strengthen transparency and public accountability in how longitudinal value is represented. Governance, monitoring, and AI-enabled risk detection infrastructure that supports prevention and continuity should be recognized as core components of clinical care delivery—not treated as administrative overhead. [PFS-STAT-1] [PFS-RVU-1]

3. Treat validated AI systems as allowable clinical infrastructure within risk-bearing models. Risk-bearing entities are structurally positioned to capture deflationary value and capacity preservation. HHS should clarify that validated AI-enabled monitoring and coordination systems qualify as allowable clinical infrastructure investments within accountable care and managed care models.

Applicable authorities: 42 CFR part 425; 42 CFR 438.242. [MSSP-1] [MMC-1]

4. Strengthen “AI-ready interoperability,” including provenance, traceability, auditability, and secure version control.

Accountable AI requires more than data exchange; it requires verifiable provenance, traceability of outputs, audit logs, version control, and secure access controls. ONC’s Predictive DSI criterion establishes source attribute transparency and intervention risk management as enforceable foundations. HHS should expand operational audit and lifecycle expectations for AI systems influencing care.

Applicable authorities: 45 CFR 170.213; 45 CFR 170.215; 45 CFR 170.315(b)(11). [ONC-USCDI-1] [ONC-API-1] [ONC-DSI-1]

5. Support computable, revocable permissions aligned with patient-directed exchange. CMS interoperability policy emphasizes patient access and patient-directed data exchange via APIs. HHS should promote computable, machine-readable, and revocable permission frameworks that allow individuals to authorize longitudinal AI-enabled systems without transaction-by-transaction consent processes that create unusable friction.

Applicable authorities: 45 CFR 164.524; 45 CFR 164.506; 42 CFR 422.119. [HIPAA-ACCESS-1] [HIPAA-TPO-1] [CMS-API-1]

6. Clarify HIPAA guidance for AI-enabled clinical operations while preserving purpose fidelity.

HHS should issue updated guidance clarifying permissible uses of protected health information (PHI) for AI-enabled care delivery, quality improvement, and safety monitoring within treatment, payment, and operations boundaries. Clear delineation of secondary use limitations would reduce uncertainty while preserving privacy protections.

Applicable authorities: 45 CFR part 160; 45 CFR 164.506; 45 CFR 164.524. [HIPAA-160-1] [HIPAA-TPO-1] [HIPAA-ACCESS-1]

7. Establish baseline governance expectations for non-medical-device AI used in clinical care. Many AI systems influencing care fall outside medical device pathways. HHS should articulate baseline expectations for non-device AI, including system-level oversight rather than per-output review; defined monitoring and escalation thresholds; auditability; version control; cybersecurity safeguards; and clear allocation of accountability. [NIST-1] [FDA-GMLP-1]

8. Adopt explicit anti-extraction and anti-capture safeguards.

Interoperability policy may be undermined if actors preserve control through lock-in, selective access, or friction-based barriers. HHS should reinforce information blocking

prohibitions and certification requirements to discourage practices that interfere with access, exchange, or use of electronic health information except as permitted under defined exceptions.

Applicable authorities: 45 CFR part 171; 45 CFR part 170. [IB-1] [ONC-USCDI-1]

9. Address liability and responsibility tensions in human–AI decision-making.

Uncertainty regarding responsibility in AI-influenced care may suppress adoption or incentivize defensive documentation rather than safety improvement. HHS should convene stakeholders to develop predictable, patient-centered harm resolution pathways, including standardized reporting frameworks, shared learning mechanisms, and exploration of administrative compensation models—while preserving due process and avoiding concealment incentives.

As AI systems operate at varying levels of autonomy, responsibility should align with degree of control and clinical involvement. When individuals independently use AI tools outside formal clinical workflows, traditional medical liability frameworks operate differently than when clinicians are directly engaged in risk-tiered oversight. HHS should support development of clearer responsibility allocation standards based on system design, escalation pathways, and documented clinician involvement—so that accountability follows governance structure rather than defaulting to defensive documentation practices.

Applicable authorities (demonstration/program mechanisms and privacy baseline): 42 U.S.C. 1315a; 42 CFR part 425; 45 CFR parts 160 and 164. [CMMI-1] [MSSP-1] [HIPAA-160-1]

QUESTION 3

For non-medical devices, what novel legal and implementation issues exist and what role should HHS play?

Non-medical-device AI used in clinical operations raises governance and implementation issues distinct from traditional medical device regulation. These systems influence prioritization, documentation, routing, and decision support without fitting neatly within existing device oversight frameworks. Key issues include:

Accountability diffusion. When AI influences prioritization, documentation, routing, triage, or recommendations, responsibility may be distributed across developers, deployers, clinicians, and institutions. ONC’s Predictive DSI criterion establishes transparency and intervention risk management expectations within certified health IT. HHS should extend this logic through guidance clarifying role delineation, documentation standards, monitoring obligations, and escalation expectations for non-device AI deployed in clinical workflows. [ONC-DSI-1]

Contracting and indemnification friction. Health systems and vendors frequently face uncertainty regarding audit rights, update control, data access, responsibility allocation, and indemnification. These ambiguities increase transaction costs and slow responsible deployment. HHS can reduce friction by publishing model contract principles and baseline expectations for logging, version control, auditability, incident reporting, and cybersecurity safeguards. [NIST-1]

Oversight models that do not scale. Requiring per-output human review for all AI-influenced decisions creates unusable friction and may restrict access without proportionate safety benefit. HHS should explicitly endorse risk-tiered monitoring frameworks that define escalation thresholds, retrospective audit mechanisms, and system-level oversight rather than transaction-by-transaction review. [NIST-1]

Versioning and change management. Non-device AI systems evolve through software updates, retraining, parameter shifts, and configuration changes. Lifecycle governance implies clear

expectations for change control documentation, performance monitoring, drift detection, impact assessment, and security validation across the total product lifecycle. HHS should articulate baseline expectations for version transparency and post-deployment monitoring. [NIST-1] [FDA-GMLP-1]

Privacy, trust, and patient rights. HIPAA establishes enforceable rights of access and defined permissions for treatment, payment, and operations. However, AI-enabled clinical operations—particularly longitudinal and agentic systems—introduce new complexity regarding durable control, purpose fidelity, secondary use boundaries, and auditability of protected health information across time and contexts. HHS should clarify how existing privacy authorities apply to AI-mediated workflows while reinforcing the principle that individuals retain ongoing, enforceable rights to access, direct, and revoke use of their health information across longitudinal systems. Governance frameworks should ensure that permissions are computable, revocable, and traceable; that AI-related data flows are auditable; and that interoperable longitudinal records reduce redundant data capture while improving data integrity.

Applicable authorities: 45 CFR parts 160 and 164; 45 CFR part 170; 45 CFR part 171. [HIPAA-160-1] [HIPAA-TPO-1] [ONC-USCDI-1] [IB-1]

QUESTION 4

For non-medical devices, what evaluation methods and metrics are most promising pre- and post-deployment? Should HHS support these processes?

Evaluation for non-medical-device AI should move beyond static pre-deployment checks toward risk-tiered, lifecycle evaluation that measures real-world outcomes, workflow effects, equity impact, and system behavior over time. [NIST-1]

Pre-deployment evaluation (risk-tiered):

1. Context-specific validation in intended clinical environments, including prospective testing where feasible, rather than reliance solely on retrospective datasets.
2. Silent-mode (“shadow”) testing to assess reliability, calibration, bias, and workflow fit without influencing live clinical decisions.
3. Human factors evaluation, including cognitive load, automation bias, interpretability, alert fatigue, and usability.
4. Equity-stratified performance analysis across demographic, socioeconomic, geographic, and clinical subgroups.

Post-deployment evaluation (continuous, lifecycle-based): 5. Automated monitoring for performance drift, safety signals, equity drift, and unintended workflow consequences. 6. Structured audit logs that enable retrospective analysis, reproducibility, and harm review. 7. Version transparency with post-update monitoring and documented change management. 8. Defined escalation thresholds that engage clinicians “on the loop” when risk profiles change or uncertainty exceeds predefined bounds. [NIST-1]

Evaluation should include metrics that extend beyond model accuracy to encompass impact on utilization, time to intervention, preserved clinical capacity, documentation burden, patient experience, and access. HHS should support these processes through grants, cooperative agreements, shared monitoring infrastructure, and multi-site learning networks that generate implementation science and reproducible evaluation toolkits. AHRQ’s CDSiC provides a model for incorporating clinicians, patients, and advocates into evaluation design and resource development. [AHRQ-CDSiC-1]

QUESTION 5

How can HHS best support private sector accreditation, certification, industry-driven testing, and credentialing?

HHS should support private sector accreditation and credentialing models that enable trust at scale while avoiding unnecessary rigidity, market concentration, or barriers to innovation.

Strengthen scalable accreditation without creating capture points. Private sector accreditation should align with baseline lifecycle governance expectations—continuous monitoring, auditability, version transparency, equity reporting, and cybersecurity safeguards—rather than rely solely on static, one-time certification that may become outdated as systems evolve. HHS can articulate minimum governance expectations while allowing industry bodies flexibility in implementation.

[NIST-1]

Prevent incumbent capture and exclusion. Assurance frameworks should promote transparency and comparability of claims and avoid proprietary or cost-prohibitive requirements that entrench market power or exclude smaller or innovative entrants. HHS can support open standards and comparability benchmarks to reduce information asymmetry without prescribing specific vendors or technical architectures. [IB-1]

Align with existing ONC governance structures. ONC's Predictive DSI transparency and intervention risk management requirements provide a structured baseline for developer-facing disclosure and risk analysis. HHS should encourage accreditation bodies to incorporate these principles to reduce duplication and promote coherence across regulatory and private assurance ecosystems. [ONC-DSI-1]

Strengthen workforce competency rather than expand licensure. Credentialing should focus on defined competency tiers rather than creation of new licensure categories. Baseline clinician education should include understanding model limitations, automation bias, calibration and uncertainty, escalation thresholds, documentation implications, and privacy and security fundamentals. Advanced competencies should prepare designated clinical AI stewards or oversight leads in lifecycle monitoring, performance and equity drift detection, incident review, human factors integration, version change management, and governance design. [NIST-1]

HHS should also support development of patient-facing AI literacy resources aligned with clinical governance frameworks. As individuals increasingly use AI tools to seek health information, clear guidance on appropriate use, limitations, escalation triggers, privacy considerations, and integration with clinical workflows will be essential. [ECRI-1]

HHS should avoid creating new licensure barriers that fragment practice across states and should study how state-level variability in scope of practice, licensure, and practice rules affects continuity and access.

Applicable authorities: 45 CFR part 170; 45 CFR part 171; 42 U.S.C. 299b-33. [ONC-USCDI-1] [IB-1] [AHRQ-AUTH-1]

QUESTION 6

Where have AI tools deployed in clinical care met or exceeded performance and cost expectations and where have they fallen short? What kinds of novel AI tools would have the greatest potential?

Where AI has met or exceeded expectations:

AI has performed most consistently where tasks are bounded, data are structured, workflows are stable, and accountability remains clear—such as selected imaging analysis, signal detection, structured classification, and narrowly defined prediction tasks. Systematic reviews demonstrate

strong performance under controlled study conditions, while also noting reporting limitations, dataset bias, and challenges in generalizability. [DL-IMAGING-1]

Where AI has fallen short:

AI has struggled when deployed as isolated point solutions within fragmented care environments lacking longitudinal context, interoperability, and clear oversight. Performance often degrades when models are introduced into complex workflows without defined escalation pathways or lifecycle monitoring. Evidence regarding generative AI suggests variable real-world performance; meta-analytic comparisons indicate that aggregate performance is not reliably superior to clinicians and may underperform expert physicians in complex cases. [TAKITA-1]

What is needed next:

The greatest potential lies not in increasingly autonomous point tools, but in integrated, longitudinal, and governed AI systems that support continuity, prevention, early risk detection, care coordination, and administrative simplification across settings. Novel AI architectures should emphasize: • Longitudinal context aggregation rather than encounter-level inference. • Risk stratification and escalation design embedded in workflow. • Automated monitoring for performance, safety, and equity drift. • Interoperable, auditable data flows aligned with patient-directed permissions. • Infrastructure that reduces unusable friction rather than adding cognitive burden. [NIST-1]

WHO guidance on large multi-modal models for health underscores systemic risks and governance requirements, reinforcing that AI should be treated as accountable clinical infrastructure within coordinated systems—not as autonomous decision-makers detached from oversight. [WHO-LMM-1]

QUESTION 7

Which role(s), decision maker(s), or governing bodies within health care organizations have the most influence on the adoption of AI for clinical care? What are the primary administrative hurdles to the adoption of AI in clinical care?

Adoption decisions for clinical AI are distributed across clinical leadership, information technology and cybersecurity, compliance and legal teams, finance, risk governance committees, and operational management. Multidisciplinary review is appropriate for safety and accountability; however, in the absence of standardized evaluation frameworks and contracting templates, decision-making can become duplicative, inconsistent, and slow.

Primary administrative hurdles include integration burden; liability uncertainty; lack of standardized lifecycle evaluation models; limited internal capacity for monitoring and drift detection; privacy and security interpretation variability under HIPAA; and information-sharing friction resulting from uneven interoperability implementation. [ONC-USCDI-1] [HIPAA-160-1]

HHS can reduce these hurdles by articulating baseline governance expectations (including monitoring, auditability, version transparency, and escalation design); clarifying HIPAA guidance for AI-enabled clinical operations; reinforcing interoperable exchange requirements; and using information blocking enforcement mechanisms where appropriate to reduce friction that impedes lawful access and reuse of electronic health information. [ONC-DSI-1] [IB-1] [HIPAA-TPO-1]

Separately, payment and valuation architecture materially influences adoption incentives. Current valuation methodologies are oriented toward discrete services and may not fully reflect longitudinal stabilization, prevention, preserved capacity, or the contributions of the broader care workforce, including nursing and advanced practice roles. HHS should strengthen transparency and public accountability in valuation processes to ensure that payment design supports continuity,

workforce sustainability, and responsible AI deployment rather than reinforcing administrative throughput incentives. [PFS-STAT-1] [PFS-RVU-1]

QUESTION 8

Where would enhanced interoperability widen market opportunities, fuel research, and accelerate the development of AI for clinical care? Consider data types, data standards, and benchmarking tools.

Enhanced interoperability would widen markets and accelerate AI development most where it enables longitudinal, permissioned, and accountable use of health information across settings. The objective is not merely data transport, but continuity of clinically relevant context and patient-directed permissions—enabling portable, risk-tiered support capabilities that operate across organizations without bespoke integrations at each site. [ONC-USCDI-1] [ONC-API-1]

Priority interoperability capabilities include:

- Longitudinal clinical context: structured and computable access to problem lists, medications, allergies, laboratory results, imaging reports, vital signs, notes metadata, and care plans, aligned with USCDI and exchanged via FHIR-based APIs. [ONC-USCDI-1] [ONC-API-1]

- Provenance and traceability: standardized fields for data source, timestamp, authoring system, version identifiers, and transformation history for both data inputs and AI-generated outputs used in clinical care. [ONC-DSI-1]

- Auditability and accountability: tamper-evident logs for access, inference events, and output use (e.g., whether an output was displayed, accepted, modified, overridden, or ignored), enabling retrospective review and quality improvement. [ONC-DSI-1]

- Computable, revocable permissions: machine-readable permissioning frameworks that support patient-directed exchange, revocation, emergency access pathways with strict auditing, and culturally or tribally aligned data stewardship requirements when applicable. [HIPAA-ACCESS-1] [CMS-API-1]

- Benchmarking and monitoring toolkits: shared post-deployment monitoring specifications—including performance drift, safety signals, equity effects, and workload impact—that allow comparison across systems without imposing proprietary reporting formats. [NIST-1]

Key federal building blocks include:

- USCDI content requirements (45 CFR 170.213). [ONC-USCDI-1]

- API standards, including FHIR (45 CFR 170.215). [ONC-API-1]

- ONC Predictive DSI transparency and intervention risk management requirements (45 CFR 170.315(b)(11)). [ONC-DSI-1]

- CMS interoperability rules expanding patient access and prior authorization APIs (including CMS Interoperability and Patient Access and the Interoperability and Prior Authorization final rule). [CMS-IPA-1] [CMS-PA-1]

- TEFCA as a nationwide interoperability framework intended to provide a floor for exchange across proprietary boundaries. [TEFCA-1]

Benchmarking should prioritize post-deployment performance, safety and equity monitoring, auditability, and real-world workflow impact—not retrospective accuracy alone. Interoperability that enables portable monitoring artifacts (not just portable data) will widen markets without sacrificing safety, accountability, and public trust. [NIST-1]

QUESTION 9

What challenges within health care do patients and caregivers wish to see addressed by the

adoption and use of AI in clinical care? Equally, what concerns do patients and caregivers have related to the adoption and use of AI in clinical care?

Patients and caregivers want continuity, timely access, reduced administrative burden, and clearer explanations aligned with their goals and circumstances. Patient-centered clinical decision support work highlights the importance of integrating patient needs, values, and preferences into decision support, moving beyond clinician-only point-of-care tools. [AHRQ-CDSIC-1]

Challenges patients and caregivers want AI to address (real-world burdens): • Navigation across fragmented systems: referrals, follow-up, prior authorization, and handoffs. [CMS-PA-1]

- Continuity of the plan: preserving the clinical “story” across time, settings, and clinicians. [ONC-USCDI-1]

- Administrative overload on households: forms, scheduling, billing confusion, and repeated history-taking. [NAM-1]

- Access constraints: long waits, geographic barriers, and limited after-hours guidance. [FR-1]

- Bias and variability: inconsistent standards of care across locations and institutions.

Core concerns (trust and agency): • Loss of agency and consent fatigue if systems require constant micro-permissions or apply opaque defaults. [HIPAA-ACCESS-1]

- Opaque decision-making and inability to see why a recommendation was made or what data drove it. [ONC-DSI-1]

- Secondary use of data that exceeds the purpose of care without meaningful transparency or patient-directed control. [HIPAA-TPO-1]

- Equity risks: performance gaps, differential access, and drift that harms under-resourced groups. [NIST-1]

- Safety failures that shift burden back onto patients and clinicians through denials, delays, or automated gatekeeping. [ECRI-1]

These concerns are intensified by the current burnout environment. When clinicians are overloaded, the health system offloads work to patients and families—who become the de facto coordinators of care. AI that merely adds new portals, messages, or alerts will worsen this. The standard should be measurable burden reduction: fewer repeated steps, fewer avoidable calls and forms, fewer untracked handoffs, and clearer accountability when something fails. [NAM-1]

QUESTION 10

Are there specific areas of AI research that HHS should prioritize to accelerate the adoption of AI as part of clinical care?

Research priority: implementation science and post-deployment system behavior.

HHS should prioritize research that evaluates AI as deployed infrastructure rather than as isolated models. This includes workflow integration, lifecycle monitoring methods, decision support governance, equity effects, audit log standards, and incident response. AHRQ’s CDSiC and patient-centered CDS initiatives provide relevant infrastructure and precedent for patient-inclusive work. [AHRQ-CDSIC-1]

Published findings (clinician error vs AI): a careful synthesis.

The literature does not support a simple claim that “AI is broadly safer than physicians” across all contexts. It supports a more precise statement: humans and AI fail differently, and outcomes depend on integration and governance. Baseline diagnostic error burden is non-trivial (NASEM’s conservative estimate), and AI tools can perform well in constrained settings. However, randomized evidence suggests clinician access to an LLM does not automatically improve diagnostic reasoning, and meta-analytic evidence suggests generative AI is not superior to expert physicians in diagnostic tasks in aggregate. Because the pace of change is significant, periodic

evaluation as AI evolves toward more agentic models is necessary. [NASEM-1] [GOH-1] [TAKITA-1] [WHO-LMM-1]

Economic and transfer analyses.

HHS should prioritize research that measures costs and transfers: who pays for implementation and monitoring; who captures savings; how AI changes clinician workload; and how reimbursement structures shape diffusion. Incentives differ across risk-bearing models and legacy payment systems can impede access to high-value AI. [FR-1]

Recommendation to HHS (human–AI complementarity):

The evidence suggests that access to an LLM does not automatically improve clinicians’ diagnostic reasoning, even when the model alone performs well in controlled evaluation. This implies that the binding constraint is not model capability but the design of human–AI collaboration—training, workflow integration, calibration, and accountability. HHS should therefore prioritize research and guidance that identifies and operationalizes the distinct elements of human involvement that confer safety and clinical value, including (1) value-laden judgment and shared decision-making, (2) accountability-bearing escalation and override decisions in high-risk contexts, and (3) verification behaviors that mitigate automation bias. These “human advantage functions” should be embedded into risk-tiered lifecycle evaluation requirements and workforce readiness initiatives so human judgment is applied where it measurably reduces harm rather than being imposed transaction-by-transaction in ways that create unusable friction and reduce access. [NIST-1] [GOH-1]

HHS could convene a multi-stakeholder working group to publish a standardized human–AI teaming framework and evaluation toolkit, aligned with lifecycle risk management approaches, usable by CMS in payment demonstrations and by ASTP/ONC in governance expectations for clinical AI. [FR-1] [NIST-1]

TABLE A: RECOMMENDATION-TO-AUTHORITY MAPPING

Recommendation (summary) | Primary CFR / program authority | HHS lever(s) | Notes
Reimbursable AI evaluation/implementation pathway (silent-mode testing, workflow evaluation, training, monitoring, equity assessment) | 42 CFR part 410; 42 CFR part 414; 42 U.S.C. 1395w-4; care management precedents | Reimbursement | Use a time-limited pathway; require auditable reporting and monitoring.

Risk-tiered lifecycle evaluation expectations for non-device clinical AI | NIST AI RMF 1.0; FDA/IMDRF GMLP guiding principles | Regulation + R&D | Encourage scalable monitoring and escalation rather than per-output review.

“AI-ready interoperability”: provenance, audit logs, versioning for AI outputs used in care | 45 CFR 170.315(b)(11) | Regulation | Build on certified health IT requirements; require traceability.

Content and API floors for longitudinal data access | 45 CFR 170.213; 45 CFR 170.215 |

Regulation | Make longitudinal context feasible across systems.

Computable, revocable permissions aligned to patient-directed exchange | 45 CFR 164.524; 45 CFR 164.506; 42 CFR 422.119 | Regulation + Reimbursement | Treat permissioning as governance infrastructure.

Reduce friction-based control; enforce information blocking constraints | 45 CFR part 171 |

Regulation | Use exceptions for privacy/security appropriately while discouraging strategic blocking.

Allowable AI infrastructure in risk-bearing models (ACO/MA/Medicaid managed care) | 42 CFR part 425; 42 CFR 438.242 | Reimbursement + R&D | Align infrastructure investments to outcomes and monitoring.

Privacy/security governance for AI-enabled operations | 45 CFR part 160; 45 CFR part 164 |

Regulation | Clarify permissible operational use vs secondary use; emphasize auditability.
 Patient-centered CDS design and evaluation | AHRQ CDSiC and patient-centered CDS resources |
 R&D | Build learning networks and shared toolkits to reduce duplication.
 Safe non-device AI governance baseline (monitoring, escalation, incident response) | 45 CFR
 170.315(b)(11); FDA/IMDRF lifecycle framing; ECRI hazard communications | Regulation +
 R&D | Standardize controls and boundary testing; address real-world misuse.
 Tribal partnership and data stewardship commitments in pilots | HHS Tribal Consultation Policy;
 IHS TEC data sharing guidance | R&D + Regulation | Co-design permissions and governance;
 respect sovereignty and Tribal Epidemiology Center roles.

TABLE B: STAKEHOLDER RECRUITMENT

Stakeholder	Role	Specific ask	Timeline (eight-week arc)
Delivery systems (academic + community)	Pilot sites; workflow integration; safety governance Participate in phased rollout; appoint governance and clinical safety leads; enable monitoring		Weeks 2–4
Employer/purchasers	Population access; benefits integration; evaluation of utilization and access Identify eligible population; agree to parallel evaluation; support claims-based endpoints		Weeks 3–6
Patients/caregivers	Health Principal governance; consent and trust design Co-design permissioning and transparency; define acceptable use boundaries		Weeks 1–8
Tribal partners / TECs / IHS-aligned stakeholders	Sovereignty-respecting governance; community priorities Establish consultation path; define data stewardship and permissions; co-author governance terms		Weeks 1–6
Health IT vendors / EHR/API implementers	Interoperability; provenance/audit schema Implement minimal provenance + audit logs; support FHIR/USCDI exchange		Weeks 2–5
Privacy/security council	HIPAA/OCR alignment; incident response Draft purpose-fidelity policy; define breach/incident pathways; review audit logging		Weeks 1–4
Human factors and safety experts	Evaluate automation bias; usability; risk controls Design human factors protocol; define escalation thresholds and rollback triggers		Weeks 2–6
Independent evaluators (academic methodologists)	Study design and analysis Pre-specify endpoints; power calculations; reporting templates		Weeks 2–5
AHRQ-style learning network partners	Shared toolkit dissemination Convert pilot learnings into reusable templates and monitoring toolkits		Weeks 6–8

ONE-PAGE PILOT INVITATION (ATTACHMENT B APPENDIX)

Health 4.0 Alliance | H4 Alliance Trust invites delivery systems, employer purchasers, patient organizations, and tribal partners to explore a multi-site, governed longitudinal AI support pilot (non-branded). The purpose is to evaluate whether risk-tiered, lifecycle-governed AI can improve continuity, access, safety, clinician burden, and equity—while preserving patient-directed permissions and auditability. [FR-1]

Pilot principles: • Health Principal-centered permissions and transparency (lifespan framing, not episodic-only). [HIPAA-ACCESS-1]

• Risk-tiered lifecycle governance, monitoring, and escalation (“on the loop” clinician oversight). [NIST-1]

• AI-ready interoperability: USCDI/FHIR access plus provenance and audit logging for outputs used in care. [ONC-USCDI-1] [ONC-API-1] [ONC-DSI-1]

• Anti-extraction safeguards: no friction-based capture; purpose fidelity for data use and auditable

access patterns. [IB-1]

- Equity and cultural alignment as measurable constraints, not marketing. [NIST-1]
- Tribal sovereignty and data stewardship co-design where tribal participation occurs, aligned with HHS's tribal consultation policy posture. [HHS-TRIBAL-1]

Proposed design (illustrative; to be finalized with partners): • Phase 0 (4–6 weeks): silent-mode testing + baseline workflow/human factors evaluation + monitoring dashboard setup. [NIST-1]

- Phase 1 (8–12 weeks): controlled rollout using stepped activation by site/team; continuous monitoring; predefined escalation and rollback thresholds. [NIST-1]
- Phase 2 (6–12 months): expansion contingent on meeting safety and equity thresholds; evaluation of utilization, access, and continuity endpoints in employer and delivery-system pathways. [AHRQ-CDSIC-1]

Ethics and oversight: The pilot will follow applicable human-subjects protections under 45 CFR part 46 where relevant; participating institutions will use their IRB/QI determination processes, with appropriate consent/authorization pathways and privacy/security controls consistent with HIPAA and institutional policy. [45CFR46-1] [HIPAA-160-1]

Endpoints (core set): • Safety signals: escalation events, diagnostic delay proxies, incident reports, rollback frequency. [NASEM-1]

- Burden: clinician time, documentation burden proxies, perceived workload. [NAM-1]
- Access/continuity: time-to-appointment, follow-up completion, longitudinal plan adherence. [FR-1]
- Equity: stratified outcomes and drift monitoring consistent with Predictive DSI governance dimensions. [ONC-DSI-1]

Partner ask: Interested partners are invited to provide a pilot site lead, governance liaison, and technical integration lead to define scope, data pathways, monitoring thresholds, and reporting templates that can be shared to reduce duplication across sites. [AHRQ-CDSIC-1]

Mermaid timeline (plain text):

mermaid

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flowchart LR

A[Partner onboarding and governance setup] --> B[Phase 0: silent-mode testing (4–6 weeks)]

B --> C[Phase 1: controlled rollout (8–12 weeks)]

C --> D[Phase 2: expansion contingent on thresholds (6–12 months)]

REFERENCES (PRIMARY URLS)

[FR-1] <https://www.govinfo.gov/content/pkg/FR-2025-12-23/pdf/2025-23641.pdf>

[HHS-AI-1] <https://www.hhs.gov/sites/default/files/hhs-artificial-intelligence-strategy.pdf>

[ONC-USCDI-1] <https://www.ecfr.gov/current/title-45/subtitle-A/subchapter-D/part-170/subpart-B/section-170.213>

[ONC-API-1] <https://www.ecfr.gov/current/title-45/subtitle-A/subchapter-D/part-170/subpart-B/section-170.215>

[ONC-DSI-1] <https://www.ecfr.gov/current/title-45/subtitle-A/subchapter-D/part-170/subpart-C/section-170.315>

[IB-1] <https://www.ecfr.gov/current/title-45/subtitle-A/subchapter-D/part-171>

[HIPAA-ACCESS-1] <https://www.ecfr.gov/current/title-45/subtitle-A/subchapter-C/part-164/subpart-E/section-164.524>

[HIPAA-TPO-1] <https://www.ecfr.gov/current/title-45/subtitle-A/subchapter-C/part-164/subpart-E/section-164.506>

[HIPAA-160-1] <https://www.ecfr.gov/current/title-45/subtitle-A/subchapter-C/part-160>

[45CFR46-1] <https://www.ecfr.gov/current/title-45/subtitle-A/subchapter-A/part-46>

[CMS-API-1] <https://www.ecfr.gov/current/title-42/chapter-IV/subchapter-B/part-422/subpart-C/section-422.119>

[CMS-IPA-1] <https://www.cms.gov/priorities/burden-reduction/overview/interoperability/policies-regulations/cms-interoperability-patient-access-final-rule-cms-9115-f>

[CMS-PA-1] <https://www.cms.gov/newsroom/fact-sheets/cms-interoperability-and-prior-authorization-final-rule-cms-0057-f>

[CMS-CCM-1] <https://www.cms.gov/files/document/chroniccaremanagement.pdf>

[TEFCA-1] <https://healthit.gov/policy/tefca/>

[MSSP-1] <https://www.ecfr.gov/current/title-42/chapter-IV/subchapter-B/part-425>

[MMC-1] <https://www.ecfr.gov/current/title-42/chapter-IV/subchapter-C/part-438/subpart-D/section-438.242>

[PFS-STAT-1] <https://uscode.house.gov/view.xhtml?edition=prelim&num=0&req=granuleid%3AUSC-prelim-title42-section1395w-4>

[PFS-REG-1] <https://www.ecfr.gov/current/title-42/chapter-IV/subchapter-B/part-414>

[PFS-410-1] <https://www.ecfr.gov/current/title-42/chapter-IV/subchapter-B/part-410>

[PFS-RVU-1] <https://www.ecfr.gov/current/title-42/chapter-IV/subchapter-B/part-414/subpart-B/section-414.22>

[CMMI-1] <https://www.law.cornell.edu/uscode/text/42/1315a>

[AHRQ-AUTH-1] <https://www.law.cornell.edu/uscode/text/42/299b-33>

[NASEM-1] <https://www.nationalacademies.org/read/21794/chapter/2>

[NAM-1] <https://nam.edu/wp-content/uploads/2020/09/4.-NAM-Taking-Action-Against-Clinician-Burnout-systems-approach.pdf>

[GOH-1] <https://jamanetwork.com/journals/jamanetworkopen/fullarticle/2825395>

[TAKITA-1] <https://www.nature.com/articles/s41746-025-01543-z>

[NIST-1] <https://nvlpubs.nist.gov/nistpubs/ai/nist.ai.100-1.pdf>

[FDA-GMLP-1] <https://www.fda.gov/media/153486/download>

[IMDRF-GMLP-1] <https://www.imdrf.org/documents/good-machine-learning-practice-medical-device-development-guiding-principles>

[WHO-LMM-1] <https://www.who.int/publications/i/item/9789240084759>

[WHO-LMM-2] <https://www.who.int/news/item/18-01-2024-who-releases-ai-ethics-and-governance-guidance-for-large-multi-modal-models>

[AHRQ-CDSIC-1] <https://digital.ahrq.gov/health-it-tools-and-resources/clinical-decision-support-cds/cds-innovation-collaborative>

[ECRI-1] <https://home.ecri.org/blogs/ecri-news/misuse-of-ai-chatbots-tops-annual-list-of-health-technology-hazards>

[DL-IMAGING-1] <https://pubmed.ncbi.nlm.nih.gov/33323251/>

[HHS-TRIBAL-1] <https://www.hhs.gov/sites/default/files/iea/tribal/tribalconsultation/hhs-consultation-policy.pdf>

